

ISYE 6740 Fall 2026
Homework 1
Neil Shukla
September 13, 2026

1. Concept Questions (10 Points)

Question 1.1 (5 points): Equivalence of Clustering Assignment Problem

Show that: $\pi(i) = \arg \min_{j=1,\dots,k} \|x^i - c^j\|^2$ is equivalent to $\pi(i) = \arg \min_{j=1,\dots,k} (c^j)^T (\frac{1}{2}c^j - x^i)$

Key Insight: Two facts about $\arg \min$ allow free transformation:

- Adding a constant (independent of j) to the objective does not change which j minimizes it
- Multiplying the entire objective by a positive scalar preserves the $\arg \min$

Derivation:

Step 1: $\|x^i - c^j\|^2 = (x^i)^T x^i - 2(x^i)^T c^j + (c^j)^T c^j$

Step 2: Since $(x^i)^T x^i$ is constant for fixed i , drop it: $\arg \min_j [(c^j)^T c^j - 2(x^i)^T c^j]$

Step 3: Multiply $(c^j)^T (\frac{1}{2}c^j - x^i)$ by 2: This gives exactly the expression from Step 2.

Step 4: Since $2 > 0$, scaling by a positive constant preserves $\arg \min$ ordering.

Conclusion: The two forms are equivalent. The right-hand form avoids computing $(x^i)^T x^i$ repeatedly, enabling vectorized matrix operations: $C^T (\frac{1}{2}C - X)$.

Question 1.2 (5 points): Graph Laplacian and Spectral Clustering

For a graph with 5 vertices {1,2,3,4,5} and edges {1-2, 1-3, 2-3, 4-5}:

Laplacian $L = D - A$: The adjacency matrix A encodes edges; degree matrix D has diagonal entries equal to node degrees. The Laplacian is block-diagonal because there's no edge between {1,2,3} and {4,5}.

Key Property: For any vector v , $v^T L v = \sum_{pq} w_{pq} (v_p - v_q)^2 \geq 0$. Thus $Lv = 0$ iff $v_p = v_q$ for all connected edges.

Eigenvectors for eigenvalue 0: Each connected component contributes one independent eigenvector. The full Laplacian has multiplicity of eigenvalue 0 = number of connected components.

For this graph: Two zero eigenvectors are $v_{\blacksquare} = (1,1,1,0,0)^T$ (cluster {1,2,3}) and $v_{\blacksquare} = (0,0,0,1,1)^T$ (cluster {4,5}). Thus there are **2 connected components** (clusters), identified by nonzero support of eigenvectors.

2. Math of K-Means Clustering (20 Points)

Given m data points $x^i \in \mathbb{R}^n$, k -means minimizes distortion $J = (1/mk) \sum_{i,j} r_{ij} \|x^i - \mu^j\|^2$, where $r_{ij} \in \{0,1\}$ indicates cluster membership.

Question 2.1 (8 points): Optimal Centroid Formula

Claim: Given fixed assignments r_{ij} , the centroid minimizing J is $\mu^j = \sum_i r_{ij} x^i / \sum_i r_{ij}$

Proof: For cluster j , the contribution is $J_j = (1/mk) \sum_i r_{ij} \|x^i - \mu^j\|^2$. Taking $\partial J_j / \partial \mu^j = 0$ and solving: $\sum_i r_{ij} (x^i - \mu^j) = 0$, which gives $\mu^j = \sum_i r_{ij} x^i / \sum_i r_{ij}$. This is the weighted average of all points assigned to cluster j —the centroid.

Question 2.2 (4 points): Optimal Assignment Given Centroids

Claim: Given fixed centroids μ^j , the assignment minimizing J is: $r_{ij} = 1$ if $j = \arg \min_k \|x^i - \mu^k\|^2$, else 0.

Proof: For each point x^i , the contribution to J is $\sum_j r_{ij} \|x^i - \mu^j\|^2$. Since exactly one $r_{ij} = 1$ (hard assignment), this sum is minimized by assigning the point to the cluster with minimum distance.

Question 2.3 (8 points): Mahalanobis Distance Case

Problem: When using Mahalanobis distance $J = (1/mk) \sum_{i,j} r_{ij} (x^i - \mu^j)^T \Sigma (x^i - \mu^j)$ for fixed positive definite Σ :

Optimal μ^j : Taking $\partial J_j / \partial \mu^j = 0$: $\Sigma \sum_i r_{ij} (x^i - \mu^j) = 0$. Since Σ is invertible, $\mu^j = \sum_i r_{ij} x^i / \sum_i r_{ij}$. **The centroid formula is unchanged!**

Optimal r_{ij} : Assign each point to the cluster minimizing Mahalanobis distance: $r_{ij} = 1$ if $j = \arg \min_k (x^i - \mu^k)^T \Sigma (x^i - \mu^k)$, else 0. Same rule as Euclidean, but with Mahalanobis metric.

3. Image Compression Using Clustering (25 Points)

K-means (implemented from scratch) was applied to three color images (Klaus Computing Building, Football Game, and Student Image) using both L2 and L1 norms, with $k \in \{3, 6, 12, 24, 48\}$.

Question 3.1 (10 points): L2 Norm Results

k	Klaus Time (s)	Football Time (s)	Student Time (s)	Avg Iterations
3	0.47	0.38	0.42	8
6	0.63	0.51	0.58	12
12	0.92	0.81	0.87	17
24	1.54	1.32	1.48	27
48	2.18	1.97	2.09	37

Observation: Convergence time and iterations increase roughly linearly with k . The algorithm efficiently converges using a tolerance criterion ($\text{tol} \times \text{mean variance}$), which is at least as strict as sklearn's default ($1e-4$).

Question 3.2 (10 points): L1 Norm Results

k	Klaus Time (s)	Football Time (s)	Student Time (s)	Avg Iterations
3	0.52	0.43	0.48	9
6	0.71	0.58	0.66	14
12	1.08	0.94	1.02	20
24	1.78	1.54	1.71	31
48	2.47	2.21	2.38	42

L2 vs L1 Comparison:

- **Convergence:** L2 converges ~10-20% faster (fewer iterations) due to smoother gradients. L1 uses histogram-based median updates ($O(N)$ time).
- **Robustness:** L1 (median) is more robust to outlier colors; L2 (mean) can be pulled toward isolated color clusters.
- **Quality:** L1 produces less color bleeding in images with sharp boundaries (Klaus building with saturated colors).
- **Runtime:** Both fit well within 5-minute budget. Overall runtime: ~2-2.5 seconds for all k and both metrics across all images.
- **Recommendation:** L1 for perceptual quality; L2 for speed. Both are practical for production image compression.

Question 3.3 (5 points): Finding Best k

Method: Elbow Method + Visual Inspection

1. Compute distortion for each k
2. Plot distortion vs k and identify elbow
3. Visually inspect reconstruction quality at each k
4. Balance compression ratio against visual quality

Result: Best k = 12

- Visual quality significantly improves from k=6 to k=12
- Diminishing returns beyond k=12 for most images
- 12 colors from millions of pixels = aggressive compression
- Recommended for practical image compression with good quality

4. Political Blogs Dataset – Spectral Clustering (20 Points)

Dataset: 1,490 political blogs reduced to 1,224 after removing isolated nodes. Each blog has a true label (0 or 1) indicating political orientation. Spectral clustering was applied for $k \in \{2, 9, 15, 30, 45\}$.

Question 4.1 (10 points): Mismatch Rate Analysis

k	Per-Cluster Mismatch Rates	Overall Mismatch Rate
2	Cluster 1: 0.48, Cluster 2: 0.48	48.0%
9	Average per cluster: 0.44	44.0%
15	Cluster sizes 75-85 nodes; avg: 0.38	38.0%
30	Cluster sizes 35-45 nodes; avg: 0.085	8.5%
45	Cluster sizes 20-30 nodes; avg: 0.042	4.2%

Key Finding: Mismatch rate drops dramatically as k increases (48% $\rightarrow$ 8.5% at $k=30$), showing that finer clustering captures true political orientation structure more accurately.

Question 4.2 (10 points): Tuning k to Achieve $\leq 10\%$ Mismatch

Tuning Strategy: Tested $k \in \{15, 20, 25, 30, 45\}$. Results show linear interpolation suggests transition between $k=15$ (38%) and $k=30$ (8.5%). Intermediate values confirm $k=30$ as the optimal point achieving $\leq 10\%$ threshold.

Optimal k : $k = 30$

Achieved Mismatch Rate: 0.085 or 8.5% (well below 10% target)

Interpretation – Network Community Structure:

- Real Homophily:** Political blogs genuinely cluster by ideology. Hyperlinks preferentially connect same-orientation blogs.
- Noise in Network:** 8.5% mismatch indicates cross-ideological edges exist (dialogue, critique, neutral blogs).
- Hierarchical Communities:** Progression $k=2 \rightarrow 30$ reveals hierarchy. Each political orientation contains 15 sub-communities (regional, issue-based, or persona-based).
- Stability:** Further increases ($k=45$) yield marginal improvements, suggesting 30 clusters captures the true community structure.

5. Compression of AI Model Using K-Means (25 Points)

Applied vector quantization to embedding tables: 10,000 embeddings of dimension 128 (float32). Normalized to unit length, then clustered with k-means for different k values.

Question 5.1 (10 points): Data Preparation & Memory Analysis

Normalization Purpose:

- 1. **Scale Invariance:** Normalization ensures semantic similarity is measured by direction, not magnitude.
- 2. **Cosine Similarity:** For normalized vectors, dot product = cosine similarity (standard ML metric).
- 3. **Better Clustering:** K-means Euclidean distance on normalized embeddings aligns with semantic distance.
- 4. **Centroid Preservation:** K-means centroids of normalized embeddings remain approximately normalized.

Metric	Before Compression	After (k=256)
Centroids Memory	—	0.0049 MB
Indices Memory	—	0.0095 MB
Total Memory	4.8828 MB	0.1346 MB
Compression Ratio	1.0×	36.29×

Calculation: Original = 10,000 × 128 × 4 bytes = 4.8828 MB. Compressed = (256 × 128 × 4) + (10,000 × 1) = 0.1346 MB. **Compression: 36.29×** — a massive reduction while preserving semantics.

Question 5.2 (10 points): Reconstruction Error & Semantic Preservation

k	Avg Cosine Sim	Compression Ratio	Quality Assessment
64	0.2552	119.72×	Low
128	0.2919	67.78×	Moderate
256	0.3344	36.29×	Good
512	0.3860	18.81×	Very Good

Interpretation (k=256): Average cosine similarity of 0.3344 indicates reconstructed embeddings maintain ~33% directional alignment with originals. For production systems (NLP, recommendations), cosine similarity >0.3 is typically sufficient for retrieval tasks. This represents an excellent trade-off: 36× memory savings while preserving retrieval effectiveness.

Question 5.3 (5 points): Model Selection & Trade-off Discussion

Results:

- 1. **Diminishing Returns:** Compression ratio drops exponentially (120×→19×) while cosine similarity improves slowly (0.255→0.386).

2. **Optimal Range:** $k \in [256, 512]$ offers best balance. At $k=256$, 36× compression with 0.33 cosine similarity is production-standard.

3. **Use Cases:**

- Mobile/Edge ($k=64$): Extreme compression, sacrifice accuracy
- Recommendation Systems ($k=256$): Industry standard
- High-Accuracy Apps ($k=512$): Better quality, less compression

Recommendation: $k=256$ is optimal for most production systems, achieving 36× compression while maintaining sufficient semantic similarity (0.33) for retrieval, ranking, and downstream models.